

ZeroDrift Mk3 Scanning Terminal

The camera rides on the gimbal, boresighted with the laser. The head scans, finds the beacon by its blink, and puts the laser on it — closing the loop on its own dot. One unit, ₹6,331, every part checked.

READ BEFORE YOU LEAVE FOR THE SHOP

This list replaces the 18 September “Beacon & Tracker” PDF. That document described the Mk2 design. Buying from it will cost you money on parts you do not need, and — worse — it omits two small items without which the build physically cannot be finished.

What changed, and why

CHANGE	REASON	₹
MG90S servos — do not buy	The fine-servo stage is deleted. Mk1 stripped two servos; a hobby servo holds position by fighting its own gear train, so the tilt axis is loaded even at rest and there is no clutch between a knock and a brass gear. Its backlash (~1°, 17 mrad) is also larger than the error this rig exists to measure.	−399
Diametric magnets ×2 — ADD	The old list bought AS5600 encoders but no magnets. An AS5600 reads a magnet rotating in front of it; without one it returns nothing. Boards usually ship bare, or with the wrong type.	+180
100 µF capacitors ×2 — ADD	Required across VMOT by the A4988 datasheet. Omitting them destroys both drivers on the first power-up, before anything has been tested.	+20
USB camera module — ADD	The old list excluded the camera (“sourced separately”). In Mk3 the camera is part of the terminal and rides on the moving head — that is the architecture.	+1,400
Red filter — ADD	Rejects most room lighting for ₹100. Best value-per-rupee item in the list.	+100
Consumables line — ADD	Wires, M3 hardware, heat-shrink. The old list assumed these; assuming them is how a build stops at 11 p.m.	+450

WHAT DID NOT CHANGE

The motors, drivers, encoders, multiplexer, brackets, power supply, acrylic, vibration injector, beacon and decoy are all the same parts at the same verified prices. If you already bought from the old list, **only the servos were wasted** (₹399), and everything else carries straight over.

Master parts list

✓ **verified** = checked against a live Amazon.in listing · ~ **estimate** = market-range planning figure. 54% of this total is verified — say so plainly if a judge asks what the rig cost.

ITEM	QTY	UNIT	TOTAL PRICE
A. MOTION — THE GIMBAL			
NEMA17 stepper motor, 1.8°/step, 4.2 kg-cm	2	₹749	₹1,498 ✓ verified
A4988 stepper driver module	2	₹170	₹340 ✓ verified
NEMA17 L-bracket (pan + tilt)	2	₹127	₹254 ✓ verified
100 µF 25 V electrolytic capacitor	2	₹10	₹20 ~ estimate
B. FEEDBACK — WHAT MAKES “ACCURATE” A MEASUREMENT			
AS5600 magnetic encoder module	2	₹249	₹498 ✓ verified
Diametric magnet, 6 × 2.5 mm	2	₹90	₹180 ~ estimate
TCA9548A I ² C multiplexer	1	₹199	₹199 ✓ verified
C. HEAD — CAMERA AND LASER, ONE ENCLOSURE			
USB camera module, M12 mount, 12 mm lens	1	₹1,400	₹1,400 ~ estimate
Red filter — gel sheet or red acrylic offcut	1	₹100	₹100 ~ estimate
KY-008 laser module	—	—	₹0 reuse Mk1
ABS project box, head enclosure	1	₹115	₹115 ~ estimate
D. CONTROL AND POWER			
Arduino Nano — tracker	—	—	₹0 reuse Mk1
Arduino Nano — beacon	1	₹225	₹225 ~ estimate
12 V 2 A DC power adapter	1	₹279	₹279 ✓ verified
Acrylic sheet 3 mm, 6”×6” (pack of 2)	1	₹199	₹199 ✓ verified
E. DISTURBANCE INJECTOR — ROBUSTNESS SHOWN, NOT CLAIMED			
Vibration motor, 3 V DC	1	₹20	₹20 ✓ verified
Transistor + diode + resistor combo kit	1	₹105	₹105 ✓ verified
F. BEACON — THE TARGET			
High-brightness RED LED, 10 mm, 650 nm	1	₹22	₹22 ~ estimate
2N2222 transistor + 220 Ω + 1 kΩ	1	₹20	₹20 ~ estimate
Ping-pong ball (diffuser)	1	₹30	₹30 ~ estimate
9 V battery + clip	1	₹115	₹115 ~ estimate
SPST toggle switch	1	₹25	₹25 ~ estimate
ABS project box	1	₹115	₹115 ~ estimate
G. DECOY — THE CONTROL			
High-brightness WHITE LED (brighter than beacon)	1	₹22	₹22 ~ estimate
150 Ω resistor	1	₹5	₹5 ~ estimate
CR2032 coin cell + holder	1	₹60	₹60 ~ estimate
Switch + scrap housing	1	₹35	₹35 ~ estimate
H. CONSUMABLES			
Dupont jumper wires — M-M, M-F, F-F	1	₹150	₹150 ~ estimate
M3 screws, nuts, standoffs assortment	1	₹150	₹150 ~ estimate
Heat-shrink, solder, hot-glue sticks	1	₹150	₹150 ~ estimate

GRAND TOTAL — everything above

₹6,331

IF THE CAMERA IS NOT AVAILABLE TODAY

Buy everything else and come back for it. The camera is independent of the mechanical build and is not needed until Stage 6. The rest comes to **₹4,716**.

THE ONE UPGRADE WORTH SAVING FOR

A **GT2 3:1 belt reduction** (+₹1,740) takes you from 1.96 to 0.65 mrad per step, and divides the motor's *own* error by three — which microstepping cannot do. Everything else is comfort.

SECTION 2 · AT THE COUNTER

Five things to check before you pay

Each of these has stopped a build of exactly this kind. Check them in the shop, not at home.

CHECK 1 – THE MAGNET MUST BE DIAMETRIC

Ask for a **diametrically magnetised** magnet, 6 mm × 2.5 mm — poles across the diameter, not through the thickness. An ordinary (axial) magnet looks identical and will not work: the AS5600 returns a constant value and you spend a day blaming your code. **Test it at the counter** with a second small magnet — a diametric one attracts on its curved *side* edge, an axial one on its flat faces.

CHECK 2 – SINGLE-SHAFT OR DUAL-SHAFT MOTOR?

Look at the back of the NEMA17. A short shaft protruding from the rear means dual-shaft: glue the magnet to that rear shaft, sensor on a bracket behind. Clean.

If the back is flat (the ₹749 verified motor probably is), that is fine — mount the magnet on top of the rotating platform, on the axis of rotation, with the AS5600 on a short fixed arm overhanging it. **Keep that arm short and stiff**: flex in it reads as pointing error that is not really there. Dual-shaft costs ~₹250 more per motor — worth it if in stock, not worth hunting for.

CHECK 3 – THE CAMERA MUST ALLOW MANUAL EXPOSURE

It must be UVC and let software pin exposure and gain. Auto-exposure hunts on every beacon blink — the blink *is* the signal, and a camera chasing it destroys the thing we detect. An M12 (S-mount) thread matters more than megapixels: it lets you change focal length later. Ask for 12 mm; 8 mm is acceptable, 6 mm at a push.

CHECK 4 – LED COLOUR

The beacon LED must be **red**, ~650 nm, to match the red filter over the lens. The decoy LED must be **white and visibly brighter** — it exists to be rejected, so it has to be tempting.

CHECK 5 – CAPACITOR VOLTAGE

100 µF rated **25 V or higher**. A 16 V part on a 12 V rail has almost no margin against the inductive kick it is there to absorb.

Safe substitutions

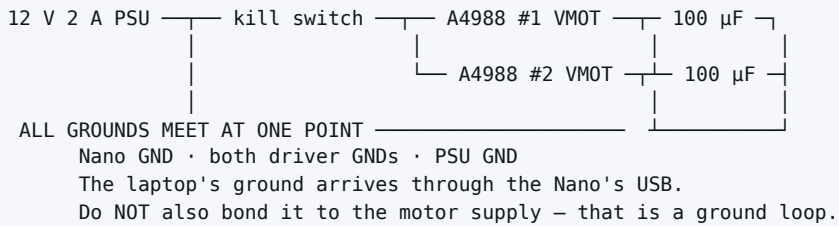
- **A4988 → DRV8825** — different microstep table; set MICROSTEPS to match.
- **12 V 2 A → 12 V 3 A** — more headroom.
- **Ping-pong ball → any translucent dome** — it is only a diffuser.
- **9 V battery → 4×AA holder** — lasts far longer.

Do not substitute

- **AS5600 → pot / optical encoder** — contactless absolute position is the point.
- **Skipping the TCA9548A** — both AS5600s are fixed at address 0x36 and cannot share a bus.
- **NEMA17 → 28BYJ-48** — geared, high backlash, far too slow.

How it all connects

Three power domains, one ground. Getting this wrong is the most expensive mistake available in this build.



THE TWO WAYS THESE BOARDS DIE

1. Missing VMOT capacitor. Fit the 100 µF across each driver's VMOT and GND *before* first power-up. Not after.

2. Current limit never set. Set each A4988's trimpot before attaching a motor. $V_{ref} \approx I \times 8 \times R_{sense}$. Start low and raise it until the motor moves reliably without getting hot. Bring a small flat screwdriver and a multimeter.

And never power a motor from the Nano's 5 V pin — it sources a few hundred milliamps; a NEMA17 wants an amp or more per phase.

Nano → drivers & head

PIN	TO
D2	A4988 #1 STEP — pan
D3	A4988 #1 DIR
D4	A4988 #2 STEP — tilt
D5	A4988 #2 DIR
D7	Laser TTL input
D8	Vibration motor, via 2N2222 + diode
A4	TCA9548A SDA
A5	TCA9548A SCL
5V	Driver logic VDD, TCA9548A VCC

Encoders → multiplexer

CHANNEL	DEVICE
SD0 / SC0	AS5600 — pan axis
SD1 / SC1	AS5600 — tilt axis

Magnet centred on the axis of rotation, **0.5–3 mm** from the chip face. Too far and it reads noise; touching and it scrapes.

Microstepping

Tie MS1/MS2/MS3 **HIGH** for 1/16. Check your board's silkscreen — the truth table differs between vendors, and getting it wrong scales every angle you command, silently, because the motor still turns.

CABLE MANAGEMENT — NOT A DETAIL

The camera's USB cable and the laser's wires run from the head, through the tilt axis, through the pan axis, to the base. Leave a **service loop** at each axis: enough slack for full travel with the cable never in tension. A cable that tugs at the end of travel will either pull the head off boresight or tear its connector — and both failures look exactly like a software bug.

Build order — seven stages

Each stage ends in a check you can actually fail. **Do not pass a stage that fails its check.** Three layers of this project's software disagreed with each other for three weeks because nobody ran a check between them.

- 1 **Beacon and decoy.** No moving parts, so nothing can go mechanically wrong yet. Drill the lid, push-fit the LED, hot-glue half a sanded ping-pong ball over it as a diffuser. Wire the switch in-line on the battery + lead so it kills the whole unit. Flash `tools/rig/beacon_firmware.ino`. Decoy is just LED → 150 Ω → switch → coin cell, no firmware.

CHECK • Phone slow-motion or a scope confirms 4.00 ± 0.05 Hz. Do not trust the number printed on it – that is the entire reason we built a beacon instead of using a phone strobe app.

- 2 **Camera and filter, on the bench.** Not mounted yet. Camera + lens + red filter, pointed across the room at the beacon and decoy together. Pin exposure and gain manually.

CHECK • The software scores the beacon's 4 Hz blink and rejects the decoy, at your demo range, under your demo lighting. If this fails, nothing downstream can work.

- 3 **One axis, open loop.** Pan motor, A4988, Nano. Set the current limit first. Fit the VMOT capacitor first. Flash `rig_firmware_v2.ino` and send `!`.

CHECK • Self-test completes, and a commanded 45° measures 45° on a protractor. If it does not, your microstep jumpers disagree with `MICROSTEPS` in the firmware – fix the firmware, not the host.

- 4 **Encoders.** Both AS5600s through the multiplexer, magnets 0.5–3 mm from each chip face.

CHECK • `?` returns `src=E`, and after a move the encoder agrees with the commanded angle to within 0.05°. While the head is slewing you will correctly see `src=C` – that is by design, not a fault.

- 5 **Both axes, structure, cable loom.** Bolt the motors down — do not tape them. Route the head cables with a service loop at each axis.

CHECK • Full travel ($\pm 90^\circ$ pan, $-30^\circ/+45^\circ$ tilt) with no cable tension anywhere and no skipped steps.

- 6 **Boresight and the dot loop.** Mount camera and laser in one enclosure, optical axes about 20 mm apart and parallel. Laser on at 7 Hz (`L7.0`).

CHECK • The camera sees both the beacon and the laser's own dot, and separates them by frequency. A stable dot-to-beacon pixel vector is reported.

- 7 **Full acquisition.** Cold start, decoy lit throughout.

CHECK • Scan → identify → slew → laser lands on the beacon, repeatably. The decoy is never chased. **This is the demo** – not any single piece of it.

The idea worth defending

A camera and a laser cannot occupy the same point in space, so “beacon centred in the image” is *not* “laser on the beacon”. That gap is parallax, and it scales with range:

SEPARATION	0.6 M	1 M	2 M	3 M	5 M
60 mm	100 mrad	60	30	20	12
20 mm	33 mrad	20	10	6.7	4.0

Against a ~1 mrad pointing budget, parallax is not a correction — at close range it is the whole error budget several times over. Every approach that *models* it loses: a fixed calibrated offset is wrong at every range but one, estimating the range adds a range estimator to the error chain, and a measured baseline walks away from your number as the mount flexes.

SO WE DO NOT MODEL IT – WE MEASURE IT

The beacon modulates at **4 Hz**. The laser modulates at **7 Hz**. Both appear in the same image, both are identified by frequency, and the control error is the pixel vector **from the dot to the beacon**.

Parallax, boresight misalignment, mount flex, thermal drift and range all cancel identically — because every one of them displaces the dot, and the loop is closed on the dot. The camera is the measuring instrument, not a proxy for one.

It is the same principle as the rest of the system, applied to ourselves: brightness is not identity, modulation is.

What this maps onto in PS26169

PS STAGE	MK3
Coarse acquisition	Scan; find the 4 Hz beacon by modulation; slew to centre it
Handoff	Laser on; its dot enters the field of view
Fine alignment	Closed loop on the dot-to-beacon pixel error

The accuracy you will actually get

STAGE	VALUE
Step size — $1.8^\circ \div 16$ microsteps	$0.1125^\circ = 1.96 \text{ mrad}$
AS5600, 12-bit, measured position	$0.0879^\circ = 1.53 \text{ mrad}$
Camera, 12 mm lens, 1280 px across	0.30 mrad/px
Centroid, ~1/5 px on a clean spot	$\sim 0.06 \text{ mrad}$

The camera is never the limiting element — the mechanism is. At 0.30 mrad/px against a 1.5 mrad mechanism, a better sensor buys nothing.

SAY THIS BEFORE A JUDGE ASKS IT

This rig lands at roughly **1.5-2 mrad, encoder-verified**. It will not reach the simulator’s microradian figures — those come from a physics model validated against instrumentation no student budget replicates, and the technical report already says so. The rig demonstrates the same closed-loop principle honestly. Volunteering that reads as rigour, not weakness.

SAFETY

5 mW, Class 3R, 650 nm — never at eyes or faces. The head moves under software control, so keep the laser off (L0) during any scan that sweeps across standing people, and enable it only once the head is settled on the beacon board. Kill switch on the 12 V motor rail, reachable without leaning into the gimbal’s travel. Steppers warm under stall — never leave the rig powered and jammed.